

108. THE ROLE OF ZONE 2 TRAINING IN ENDURANCE SPORTS

knowledgeiswatt.substack.com/p/108-the-role-of-zone-2-training-in

Knowledgeiswatt

October 15, 2025



- In [a very debated previous KIW Research Note](#), we saw that, **when matching total work (read energy expenditure), zone 2 (low intensity, below the first threshold) induces less aerobic adaptations gains (increase in VO2max and thresholds) when compared to higher exercise intensities (zone 3, 4, 5, 6).**
- However all successful endurance athletes spend about 80-90% of their total training time at low intensity, irrespectively of the endurance speciality considered: running, swimming, triathlon, rowing, cross-country skiing, road cycling.
- In addition, also some scientific studies seems to suggest that high volume low intensity training is crucial for superior endurance performance. For example, in [the last KIW Research Note](#) we saw that marathon runners with lower marathon finish time (faster runners) perform the same weekly amount of training (minutes) at medium and high intensity but a significantly higher volume of low intensity, than marathoners with higher marathon finish times (slower runners).

So, why a high volume of low intensity training (zone 2 in a 5/7 zones model) is largely used by successful endurance athletes?

A study published by Pekka Matomäki (Paavo Nurmi Centre & Unit for Health and Physical Activity, University of Turku, Finland) on *European Journal of Applied Physiology* in 2025 tried to answer this question. (1)

WHAT DID HE DO?

This scientific piece was a ***perspective***. What does it mean? No data were collected and no experiment/training protocol was performed. The author (Matomäki) 'just' tried to make some hypothesis about *why a high volume of zone 2 (low intensity) training is largely used by successful endurance athletes based on all the previously scientific published data*.

WHAT DID HE SUGGEST?

Here below you can find some of the Matomäki's different hypothesis. Of all those made by the author, I have included only those that make the most sense to me, while I have excluded those that I considered less relevant. If you want to see all Matomäki's hypothesis you can read the original full text paper, it is free access! [Full Text](#).

From here the abbreviation 'LIT' refers to low intensity training (zone 2).

HYPOTHESIS 1

"LIT PROVIDES MAINTENANCE OR SLIGHT IMPROVEMENTS TO PERFORMANCE WITHOUT CUMULATING STRESS"

In my point of view, while this Matomäki's statement can be considered a good start, I think it could be more appropriate to say:

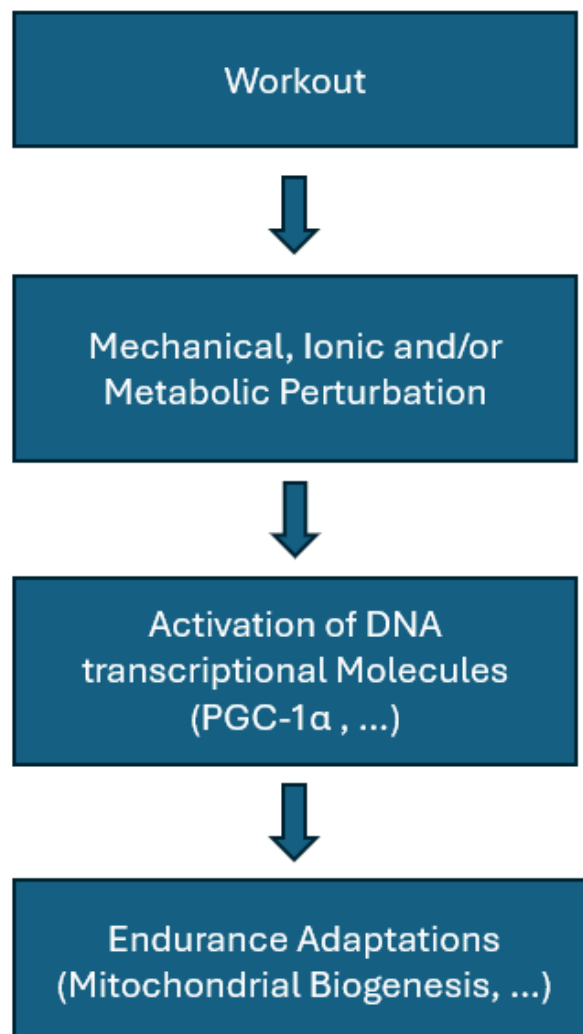
"LIT can stimulate physiological and performance adaptations, while cumulating less stress and fatigue when compared to higher exercise intensities"

In fact complete cardiac autonomic recovery (heart rate variability), which can be read as readiness to absorb a given load and successfully adapt to a training stimulus, can take 2-3 days to return at baseline level after a session with focus medium or high intensity sessions. On the other hand, low intensity training (of course, depending on the duration) seems to affect much less negatively cardiac autonomic recovery. So, **to obtain long-term performance gains, progressively increasing the training volume at low intensity to complement the more powerful but limited (2/3 times per week) amount of sessions with focus on medium and high intensity training which can be completed within each microcycle (week) is probably the best sustainable way.**

HYPOTHESIS 2

“LIT IS AN ALTERNATIVE METHOD FOR MOLECULAR ADAPTATION SIGNALS”

If you want to be a master of endurance training methodology, you need to understand how training adaptations occur. Here's how: A certain workout, or even just a particular moment within it, induces a mechanical, ionic or metabolic perturbation in certain body's physiological systems. These changes trig the activation of molecules (such as **PGC-1 α**), that stimulate the expression of certain DNA genes, which in turn make positive endurance related adaptations happen.



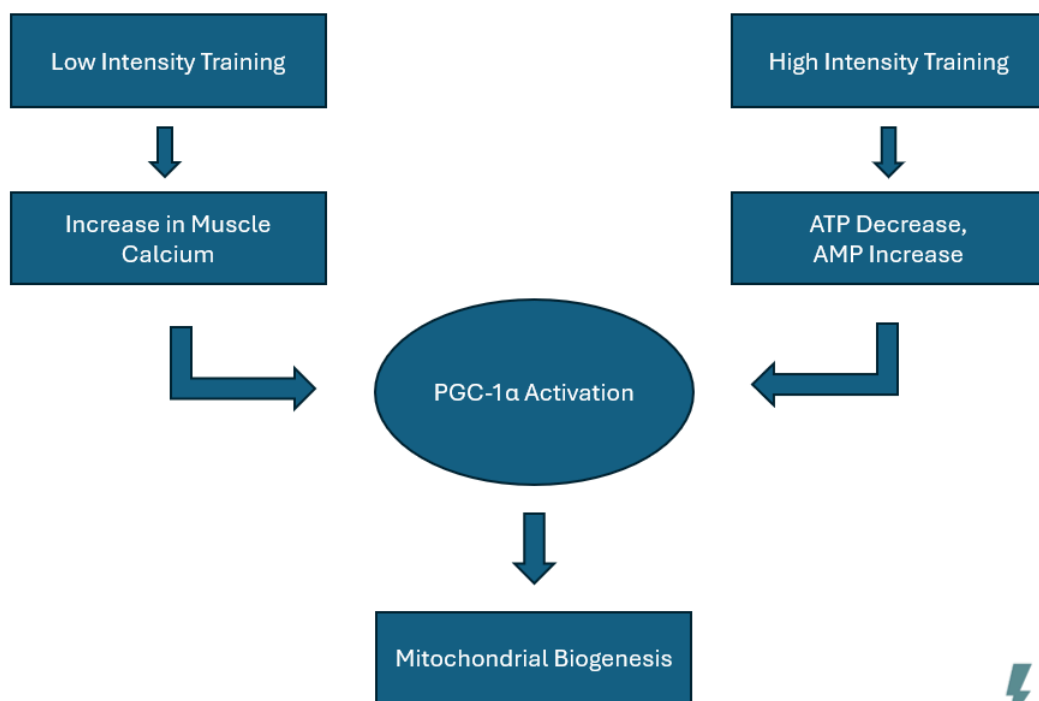
The peroxisome proliferator-activated receptor gamma coactivator 1 (**PGC-1 α**) has been described by some as the “master switch” for mitochondrial biogenesis. (2) Mitochondria are the the place where energy is produced through the oxidative pathways in the muscle cells. An increase in mitochondria means you become aerobically stronger and so higher your power at thresholds, FTP, Critical Power and VO2max.

PGC-1 α seems to be activated through various different signals:

- an increase in free radicals
- an increase in muscle calcium (Ca^{++})
- an altered energy status (i.e. lower ATP and higher AMP concentrations).

Interestingly, both low and high intensity training increase PGC-1 α content, but it seems like they act through two different pathways:

- HIT stimulates PGC-1 α through primarily metabolic pathways (lower ATP and higher AMP concentrations)
- LIT through ionic perturbation (increase in muscle calcium (Ca^{++})). Therefore low intensity training can represent an alternative adaption pathway which, introducing variability, strengthen in a complementary way the metabolic adaptive stimulus given by higher exercise intensities.



HYPOTHESIS 3

“LIT IS NEEDED PSYCHOLOGICALLY”

Both research data and experience from the field seem to suggest that high intensity training is much more psychologically demanding when compared to low intensity training. This can be another explanation about why the amount of medium and high intensity training doable within a microcycle in the long term is limited, and so it should be complemented with a progressively increased high volume of low intensity training to improve performance.

*HYPOTHESIS 4

“A HIGH VOLUME OF LIT OPTIMIZES DURABILITY”

This hypothesis have not been mentioned by Matomäki (this the reason of the “*”), but I think it’s very relevant for endurance sports with competition longer than ~2 hours, so I added it. In fact, **previously published scientific data (3) showed that a high volume of low intensity training is important to optimize durability (fatigue resistance/fatigue resilience)**, that is decreasing as less as possible the performance level during prolonged exercise.

A BRIEF SUMMARY ON LOW INTENSITY TRAINING (ZONE 2)

So, based on this and the lasts previous KIW Research Notes, here a brief summary about the role of low intensity or zone 2 (in a 5/7 zones model, below the first threshold) training in endurance sports:

- When matching total work (read energy expenditure), zone 2 (low intensity, below the first threshold) induces less aerobic adaptations gains (increase in VO2max and thresholds) when compared to higher exercise intensities (zone 3, 4, 5, 6)
- However, all successful endurance athletes spend about 80-90% of their total training time at low intensity, irrespectively of the endurance speciality considered: running, swimming, triathlon, rowing, cross-country skiing, road cycling.
- Marathon runners with lower marathon finish time (faster runners) perform the same weekly amount of training (minutes) at medium and high intensity, but a significantly higher volume of low intensity, than marathoners with higher marathon finish times (slower runners).
- So, a high volume of low intensity training seems important to optimize endurance performance.

- Why? Here some hypothesis:
 1. LIT can stimulate physiological and performance adaptations, while cumulating less stress and fatigue when compared to higher exercise intensities. So, to obtain long-term performance gains in a sustainable way, progressively increasing the training volume at low intensity to complement the more powerful but limited (2/3 times per week) amount of sessions with focus on medium and high intensity training which can be completed within each microcycle (week) is likely the best way.
 2. LIT is an alternative method for molecular adaptation signals, which means that it stimulates aerobic physiological adaptation through a different (and likely complementary) pathway from high intensity training. So, summing LIT+HIT can induce more aerobic adaptations when compared to use only LIT or only HIT.
 3. LIT is psychologically less demanding than when compared to higher exercise intensities.
 4. A high volume of LIT optimize durability (fatigue resistance).
- Based on all these points, my suggestion is that every good training program should include 2/3 weekly session with focus on medium or high intensity training. As a consequence, if you are time-crunched and training only ~5-10 hours per week, it doesn't make sense to train 80-90% of your time below your first threshold (like pro athletes do). If you trained like that, you wouldn't be optimizing your time and you would be leaving something on the table. After you've already inserted these 2/3 weekly medium or high intensity training sessions, my suggestion to further improve your performance (in a long term sustainable way) is to progressively increase the amount of low intensity training.

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Science

REFERENCES:

-
1. Matomäki P. Why low-intensity endurance training for athletes? Eur J Appl Physiol. 2025 Sep;125(9):2401-2407. doi: 10.1007/s00421-025-05843-w. Epub 2025 Jun 27.
 2. Laursen PB. Training for intense exercise performance: high-intensity or high-volume training? Scand J Med Sci Sports. 2010 Oct;20 Suppl 2:1-10.
 3. Jones AM, Kirby BS. Physiological Resilience: What Is It and How Might It Be Trained? Scand J Med Sci Sports. 2025 Mar;35(3):e70032.





[15 Likes](#)